Annular Tail Reduction to the Moving Spectral Head

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Abstract

The direct annular criterion of RH-300 was left inactive because no norm estimate was known for the actual complement-to-anchor logarithmic mismatch. We prove that its high-order part is already controlled. For every fixed $1.4 < \rho < \rho_* = 1.426787\ldots$, the noisy modulus-complement tail and the deterministic target tail beyond $m_\sigma = \lceil 4\log(1/\sigma)\rceil$ vanish in both $H^\infty(\rho)$ and $H^2(\rho)$ with explicit bounds. Consequently full annular convergence is equivalent to convergence of the moving polynomial head below m_σ. This locates the remaining obstruction without proving that the moving head converges.

1 Setup

Put

$$q = \frac{1}{2}, \qquad R = \frac{7}{5}, \qquad q_* = (0.85\lambda)^{-1} = 0.700875225854775\ldots, \qquad \rho_* = q_*^{-1}.$$

Let C_σ be the modulus-complete spectral complement from RH-282, with squared spectral mass $M_\sigma \le \sigma^{-1}$, and write

$$\tau_{\sigma,n} = \operatorname{Tr} C_\sigma^n, \qquad |\tau_{\sigma,n}| \le \sigma^{-1} q^{n-2}.$$

The deterministic numerator anchors satisfy the all-order RH-267 envelope $|a_n| < 48q_*^n$. Define

$$g_\sigma(z) = \sum_{n\ge 2} \frac{\tau_{\sigma,n} - a_n}{n} z^n, \qquad g_\sigma^{<m}(z) = \sum_{2\le n<m} \frac{\tau_{\sigma,n} - a_n}{n} z^n.$$

For $q\rho < 1$ and $q_*\rho < 1$, both series are well defined on the closed disk of radius ρ.

2 Tail theorem

Theorem 2.1 (slope-four annular tail reduction)**.** *Fix $R < \rho < \rho_*$ and put $m_\sigma = \lceil 4\log(1/\sigma)\rceil$, $x = q\rho$, and $y = q_*\rho$. Then*

$$\|g_\sigma - g_\sigma^{<m_\sigma}\|_{H^\infty(\rho)} \le \frac{4\sigma^{-1}x^{m_\sigma}}{m_\sigma(1-x)} + \frac{48y^{m_\sigma}}{m_\sigma(1-y)},$$
$$\|g_\sigma - g_\sigma^{<m_\sigma}\|_{H^2(\rho)} \le \frac{4\sigma^{-1}x^{m_\sigma}}{m_\sigma\sqrt{1-x^2}} + \frac{48y^{m_\sigma}}{m_\sigma\sqrt{1-y^2}}.$$

Both right sides tend to zero. More precisely, apart from the common $1/\log(1/\sigma)$ factor, their two powers of σ are

$$4\log(2/\rho) - 1 \quad \text{and} \quad 4\log(\rho_*/\rho).$$

Hence for $X = H^\infty(\rho)$ or $H^2(\rho)$,

$$\|g_\sigma\|_X \longrightarrow 0 \quad \Longleftrightarrow \quad \|g_\sigma^{<m_\sigma}\|_X \longrightarrow 0.$$

Proof. For the noisy tail, use $|\tau_{\sigma,n}| \le 4\sigma^{-1}q^n$ and

$$\sum_{n\ge m} \frac{x^n}{n} \le \frac{x^m}{m(1-x)}, \qquad \left(\sum_{n\ge m} \frac{x^{2n}}{n^2}\right)^{1/2} \le \frac{x^m}{m\sqrt{1-x^2}}.$$

The same estimates with $48y^n$ give the target terms. Since $m_\sigma \ge 4\log(1/\sigma)$ and $x, y < 1$, the displayed powers follow. They are positive throughout $R < \rho < \rho_*$. The equivalence is the triangle inequality applied to the vanishing tail. □

3 Numerical protocol and scope

The reproducible computation evaluates the two certified bounds at $\rho = 1.405, 1.41, 1.42$. At $\rho = 1.41$ the noisy and target exponents are $0.3982299047\ldots$ and $0.0473427916\ldots$. The small target exponent explains why finite-noise upper bounds can remain large even though the asymptotic theorem is strict.

Remark 3.1. The theorem closes only the analytic high-order tail. It supplies no estimate for the moving polynomial $g_\sigma^{<m_\sigma}$ and therefore does not activate RH-300. Gates A–E remain false/open; no Hilbert–Polya, Riemann-zero, zeta-divisor, von Mangoldt trace, or RH conclusion is made.